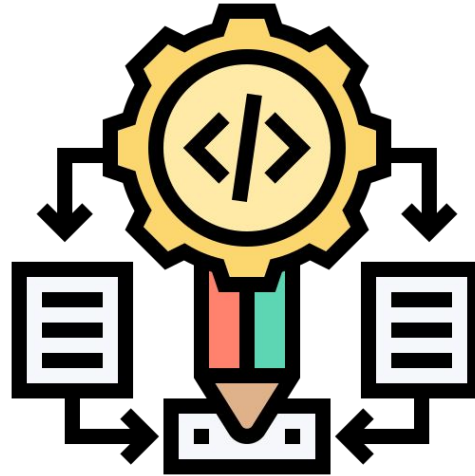


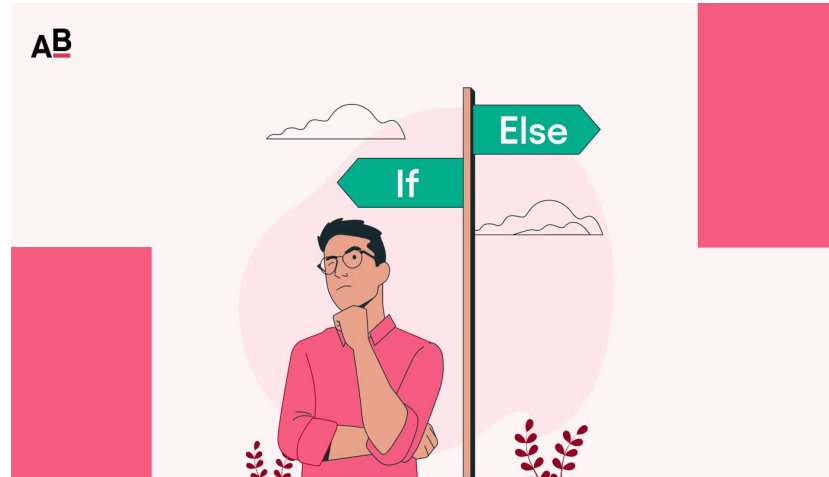
Algorithm 1



Dr Benabderrezak
Researcher in Cyber Security
Full-stack Web developer

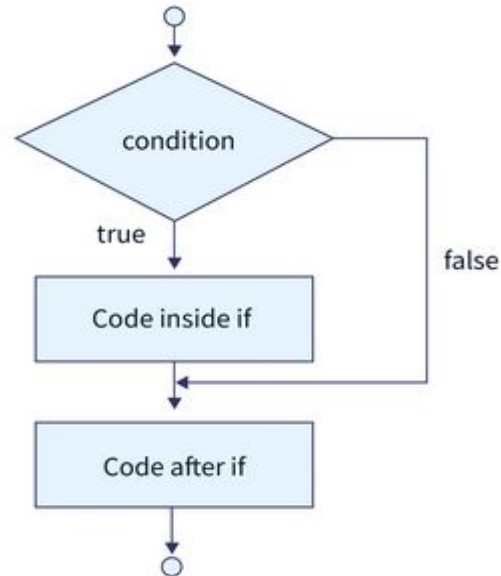
Conditions

شروط



Conditions

- Also known as **conditional statements** or **control structures**
- Used to **make decisions based on certain conditions**



Conditions

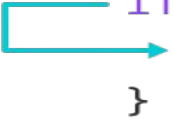
Algorithm Syntax	C Syntax
<pre>Algorithm Example Begin if x > 0 then // Do something else // Do something else End</pre>	<pre>#include <stdio.h> int main() { int x = 5; if (x > 0) { // Do something } else { // Do something else } return 0; }</pre>

Conditions - If statement

- The if statement is used to execute a block of code if a specified condition is true

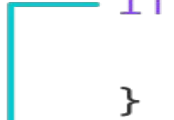
Condition is true

```
int number = 10;  
  
if (number > 0) {  
    // code  
}  
  
// code after if
```



Condition is false

```
int number = 10;  
  
if (number < 0) {  
    // code  
}  
  
// code after if
```



Conditions - If statement

```
#include <stdio.h>
void main() {
    int num = 10;

    // Check if num is greater than 0
    if (num > 0) {
        printf("%d is positive\n", num);
    }
}
```



10 is positive

Conditions - If-else statement

```
if (condition) {
```

```
    // code to execute if the condition is true
```

```
} else {
```

```
    // code to execute if the condition is false
```

```
}
```

Conditions - If-else statement

```
#include <stdio.h>
void main() {
    int num = -7;

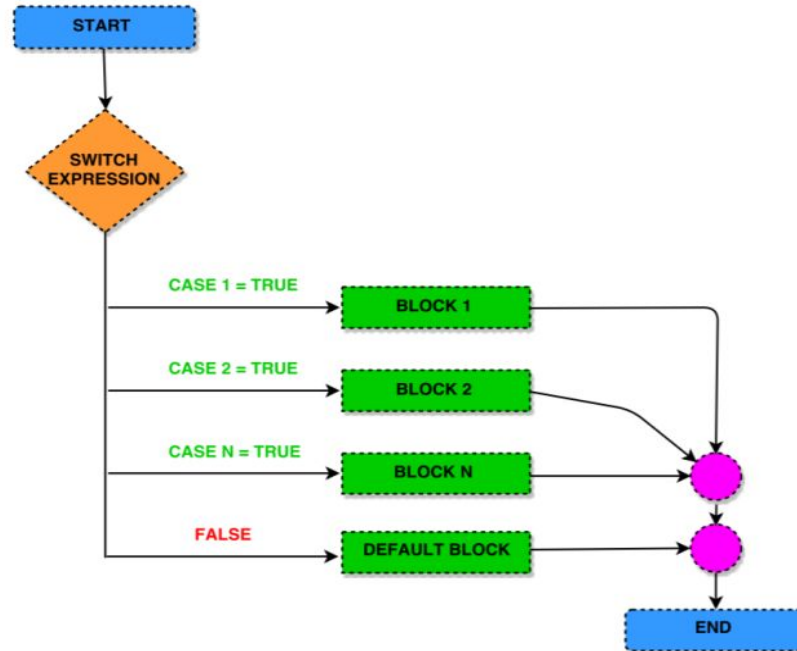
    // Check if num is greater than 0
    if (num > 0) {
        printf("%d is positive\n", num);
    }else {
        printf("%d is negative\n", num);
    }
}
```



-7 is negative

Conditions - switch statement

The switch statement is used to perform different actions based on different possible values of a variable



Conditions - switch statement

```
switch(test-expression)
{
    case value 1: statement;
                  break;
    case value 2: statement;
                  break;
    case value 3: statement;
                  break;
    default: statement;
             break;
}
```

Conditions - switch statement

```
#include <stdio.h>
void main() {
    int choice=2;

    // Start of switch statement
    switch (choice) {
        case 1: // If choice == 1
            printf("You chose number 1\n");
            // Break is used to exit the switch block
            break;

        case 2: // If choice == 2
            printf("You chose number 2\n");
            break;
        default:
            printf("Invalid choice.\n");
    }
}
```



You chose number 2



Thank
You!

